

What is Normalization?

Definition

Normalization is the process of organizing data in a database to minimize redundancy, eliminate anomalies, and improve data integrity by decomposing large tables into smaller, well-structured tables.

Imagine you have one huge table storing everything:

StudentID	StudentName	Department	HOD	Course	Grade
101	Aadya	ECE	Dr. Mehta	DSP	A
101	Aadya	ECE	Dr. Mehta	Microprocessors	A
102	Arpit	ECE	Dr. Mehta	Signals & Systems	B

Problems:

- Aadya's name repeats.
- Department repeats.
- HOD repeats.
- Wasted storage.
- Difficult to update.

Normalization breaks this into smaller logical tables.

After normalization:

Student			Department	
StudentID	StudentName	DepartmentID	DepartmentID	HOD
101	Aadya	ECE	ECE	Dr. Mehta
102	Arpit	ECE		

Enrollment		
StudentID	Course	Grade
101	DSP	A
101	Microprocessors	A
102	Signals & Systems	B

without normalization there are problems like redundancy , update anomaly , insertion anomaly , deletion anomaly ,

Objectives of Normalization

Normalization has several important objectives.

Objective 1 — Eliminate Redundancy

Avoid storing the same information repeatedly.

Objective 2 — Improve Data Integrity

Every fact should exist in exactly one place.

Example

Department name should exist only in the Department table.

Objective 3 — Remove Anomalies

Normalization removes:

- Update anomalies
- Insertion anomalies
- Deletion anomalies

Objective 4 — Maintain Consistency

If one value changes,
it changes only once.

Example

Changing the HOD requires updating only one row.

Objective 5 — Better Database Design

Smaller tables are:

- Easier to understand
- Easier to maintain
- Easier to extend

Objective 6 — Reduce Storage

Duplicate values are removed.

Objective 7 — Improve Maintainability

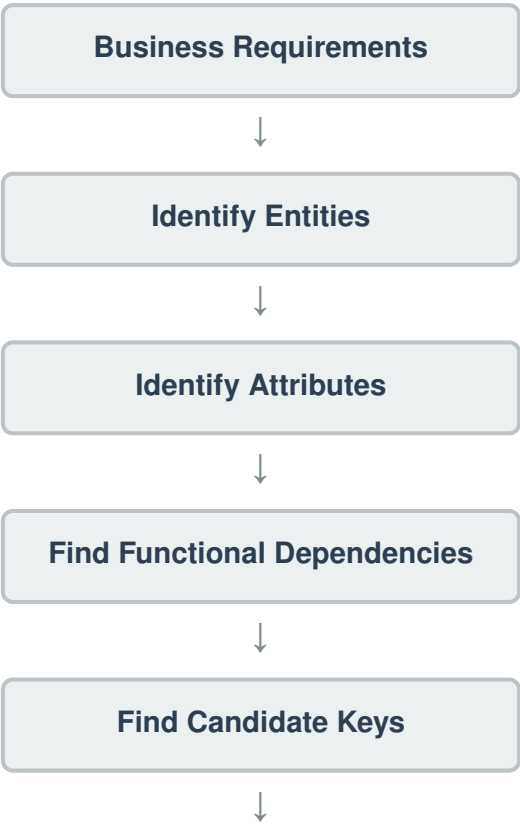
Future schema changes become simpler.

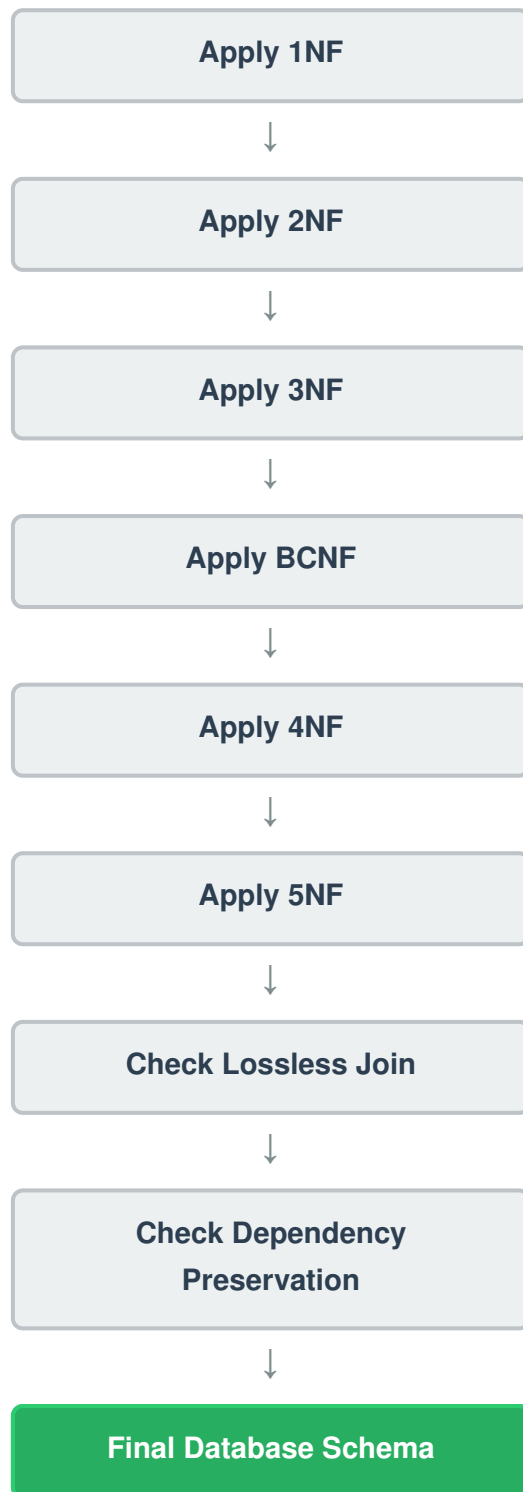
Why Multiple Normal Forms Exist.

Because each Normal Form solves a different problem.

Form	Solves
1NF	Removes: Repeating groups.
2NF	Removes: Partial Dependency.
3NF	Removes: Transitive Dependency.
BCNF	Fixes situations where 3NF is still insufficient.
4NF	Removes: Multivalued Dependency.
5NF	Removes: Join Dependency.

Complete Workflow Diagram





Is Normalization mandatory?

Answer

No.

Normalization is a design technique, not a mandatory rule.

In practice:

- OLTP systems (banking, e-commerce transactions, student management) are usually normalized because data consistency is critical.
- OLAP systems, reporting databases, and analytics platforms often use Denormalization to improve query performance.

The level of normalization depends on the application's requirements.

Does Normalization improve performance?

Answer

It depends.

- It usually improves update, insert, and delete performance because data is stored only once.
- It may reduce read performance because queries often require joins across multiple tables.